

Depth and Reach

Stress, Statistics, and the Next 40 Years

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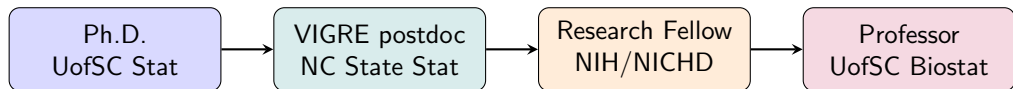
University of South Carolina

Department of Statistics

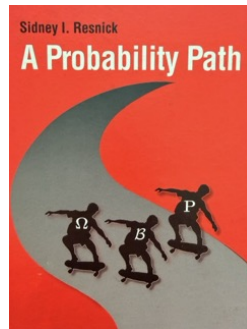
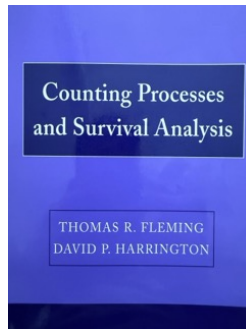
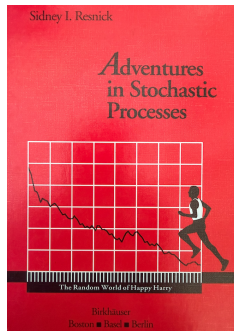
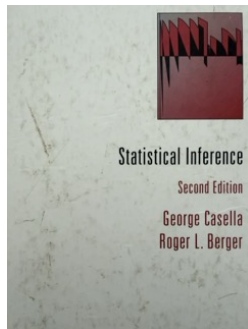
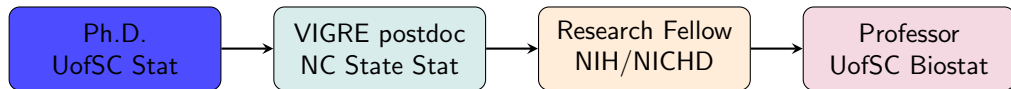
40th Anniversary Conference



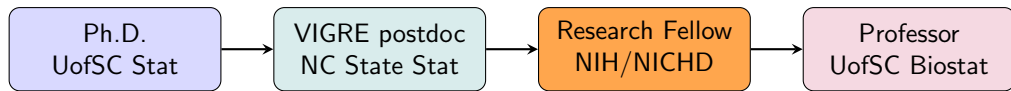
My Career Path



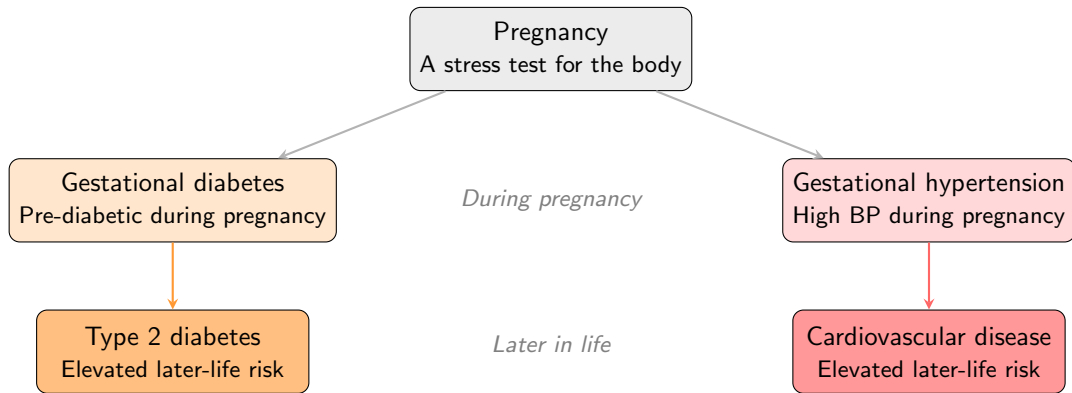
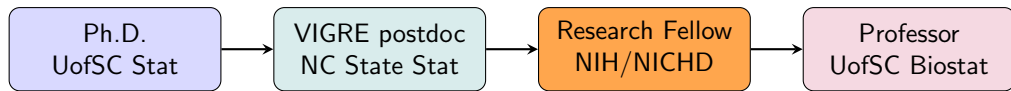
My Career Path



My Career Path



My Career Path



Stress tests reveal what was already there

- ▶ Pregnancy stress tests the body
 - ▶ Latent vulnerabilities become visible

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- ▶ COVID-19 stressed scientific infrastructure
 - ▶ Latent vulnerabilities became visible

Stress tests reveal what was already there

- ▶ Pregnancy stress tests the body
 - ▶ Latent vulnerabilities become visible
- ▶ COVID-19 stressed scientific infrastructure
 - ▶ Latent vulnerabilities became visible
- ▶ For (us) statisticians:
 - ▶ Our models, p -values, and uncertainty estimates were suddenly front page news

Statistics on the front page

FINANCIAL TIMES



Covid-19: The global crisis – in data

Charts and maps show paradoxes of a pandemic that has claimed a million lives

by FT Visual & Data Journalism team OCTOBER 18 2020



The Washington Post

Democracy Dies in Darkness



Health

Why outbreaks like coronavirus spread exponentially, and how to “flatten the curve”

By Harry Stevens March 14, 2020

FiveThirtyEight

Politics Sports Science & Health Economics Culture

Why It's So Freaking Hard To Make A Good COVID-19 Model

By [Tobias Fünfschilling](#), [Lorenz Böttcher](#) and [Tobias Fünfschilling](#)
Filed under [Economics](#)
Published Mar. 31, 2020



COVID-19 Dashboard

COVID-19 Dashboard by the Center for Systems Science and Engineering (CSE) at Johns Hopkins University (JHU)

Total Cases

100,002,578

1,589,384

Total Deaths

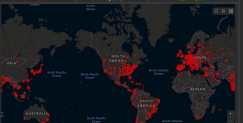
1,088,224

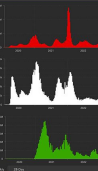
10,173

Total Positive Cases Administered

649,501,939

5,571,824





The NEW ENGLAND
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MARCH 26, 2020

VOL. 382 NO. 13

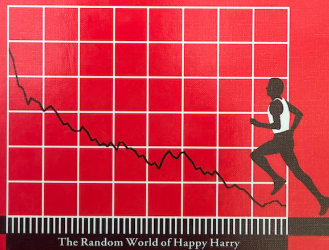
Early Transmission Dynamics in Wuhan, China,
of Novel Coronavirus–Infected Pneumonia

Qun Li, M.Med., Xuhua Guan, Ph.D., Peng Wu, Ph.D., Xiaoye Wang, M.P.H., Lei Zhou, M.Med.,
Yeqing Tong, Ph.D., Ruiqi Ren, M.Med., Kathy S.M. Leung, Ph.D., Eric H.Y. Lau, Ph.D., Jessica Y. Wong, Ph.D.,
Xuesen Xing, Ph.D., Nijuan Xiang, M.Med., Yang Wu, M.Sc., Chao Li, M.P.H., Qi Chen, M.Sc., Dan Li, M.P.H.,
Tian Liu, B.Med., Jing Zhao, M.Sc., Man Liu, M.Sc., Wenxiao Tu, M.Med., Chuding Chen, M.Sc.,
Lianmei Jin, M.Med., Rui Yang, M.Med., Qi Wang, M.P.H., Suhua Zhou, M.Med., Rui Wang, M.D.,
Hui Liu, M.Med., Yinbo Luo, M.Sc., Yuan Liu, M.Med., Ge Shao, B.Med., Huan Li, M.P.H., Zhongfa Tao, M.P.H.,
Yang Yang, M.Med., Zhiqiang Deng, M.Med., Boxi Liu, M.P.H., Zhitao Ma, M.Med., Yanping Zhang, M.Med.,
Guoqing Shi, M.P.H., Tommy T.Y. Lam, Ph.D., Joseph T. Wu, Ph.D., George F. Gao, D.Phil.,
Benjamin J. Cowling, Ph.D., Bo Yang, M.Sc., Gabriel M. Leung, M.D., and Zijian Feng, M.Med.

Statistical methods: Bayesian transmission models formulated with renewal equations to estimate the basic reproduction number R_0 .

Sidney I. Resnick

Adventures in Stochastic Processes



The Random World of Happy Harry

Birkhäuser
Boston ■ Basel ■ Berlin

Happy Harry was Back!

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DECEMBER 31, 2020

VOL. 383 NO. 27

Safety and Efficacy of the BNT162b2 mRNA Covid-19 Vaccine

Fernando P. Polack, M.D., Stephen J. Thomas, M.D., Nicholas Kitchin, M.D., Judith Absalon, M.D.,
Alejandra Gurtman, M.D., Stephen Lockhart, D.M., John L. Perez, M.D., Gonzalo Pérez Marc, M.D.,
Edson D. Moreira, M.D., Cristiano Zerbini, M.D., Ruth Bailey, B.Sc., Kena A. Swanson, Ph.D.,
Satrajit Roychoudhury, Ph.D., Kenneth Koury, Ph.D., Ping Li, Ph.D., Warren V. Kalina, Ph.D., David Cooper, Ph.D.,
Robert W. Frenc, Jr., M.D., Laura L. Hammitt, M.D., Özlem Türeci, M.D., Haylene Nell, M.D., Axel Schaefer, M.D.,
Serhat Ünal, M.D., Dina B. Tresnan, D.V.M., Ph.D., Susan Mather, M.D., Philip R. Dormitzer, M.D., Ph.D.,
Uğur Şahin, M.D., Kathrin U. Jansen, Ph.D., and William C. Gruber, M.D., for the C4591001 Clinical Trial Group*

Statistical methods: Randomized controlled trial with Bayesian success criteria;
posterior credible intervals for vaccine efficacy estimation.

Analysis

The WHO estimates of excess mortality associated with the COVID-19 pandemic

<https://doi.org/10.1038/s41586-022-05522-2>

Received: 19 May 2022

Accepted: 3 November 2022

Published online: 14 December 2022

Open access

 Check for updates

William Msemburi^{1✉}, Ariel Karlinsky², Victoria Knutson³, Serge Aleshin-Guendel³,
Somnath Chatterji¹ & Jon Wakefield^{3,4}

The World Health Organization has a mandate to compile and disseminate statistics on mortality, and we have been tracking the progression of the COVID-19 pandemic since the beginning of 2020¹. Reported statistics on COVID-19 mortality are problematic for many countries owing to variations in testing access, differential diagnostic capacity and inconsistent certification of COVID-19 as cause of death. Beyond what is directly attributable to it, the pandemic has caused extensive collateral damage that

Statistical methods: Ensemble of Poisson regression, spline, and Gaussian process models to estimate excess mortality from incomplete global data.

ORIGINAL ARTICLE

BNT162b2 mRNA Covid-19 Vaccine in a Nationwide Mass Vaccination Setting

Noa Dagan, M.D., Noam Barda, M.D., Eldad Kepten, Ph.D., Oren Miron, M.A.,
Shay Perchik, M.A., Mark A. Katz, M.D., Miguel A. Hernán, M.D.,
Marc Lipsitch, D.Phil., Ben Reis, Ph.D., and Ran D. Balicer, M.D.

ABSTRACT

BACKGROUND

Inno-
vices,
CD

As mass vaccination campaigns against coronavirus disease 2019 (Covid-19) commence worldwide, vaccine effectiveness needs to be assessed for a range of out-

Statistical methods: Observational study using propensity score matching to estimate real-world vaccine effectiveness from nationwide registry data.

Païro-Castineira et al. (2021) — Nature

nature

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Article | Published: 11 December 2020

Genetic mechanisms of critical illness in COVID-19

[Erola Païro-Castineira](#), [Sara Clohisey](#), [Lucija Klaric](#), [Andrew D. Bretherick](#), [Konrad Rawlik](#), [Dorota Pasko](#), [Susan Walker](#), [Nick Parkinson](#), [Max Head Fourman](#), [Clark D. Russell](#), [James Furniss](#), [Anne Richmond](#), [Elvina Gountouna](#), [Nicola Wrobel](#), [David Harrison](#), [Bo Wang](#), [Yang Wu](#), [Alison Meynert](#), [Fiona Griffiths](#), [Wilna Oosthuyzen](#), [Athanasios Kousathanas](#), [Loukas Moutsianas](#), [Zhijian Yang](#), [Ranran Zhai](#), [The GenOMICC Investigators](#), [The ISARIC4C Investigators](#), [The COVID-19 Human Genetics Initiative](#), [23andMe Investigators](#), [BRACOVID Investigators](#), [Gen-COVID Investigators](#), ... [J. Kenneth Baillie](#) 

[+ Show authors](#)

[Nature](#) **591**, 92–98 (2021) | [Cite this article](#)

398k Accesses | **1356** Citations | **2614** Altmetric | [Metrics](#)

Statistical methods: Genome-wide association study (GWAS) meta-analysis with genome-wide significance threshold ($p < 5 \times 10^{-8}$), Mendelian randomization, and fine-mapping to identify causal variants for COVID-19 severity.

What worked well I: early inference and forecasting

Transmission Fast descriptive inference from incomplete early data.

Interventions Semi-mechanistic models turned delays and deaths into usable estimates.

Forecasts Ensembles converted model disagreement into more reliable short-term prediction.

- ▶ When the target was clear and the horizon was short, statistics was a superpower.

What worked well II: vaccine trials

- ▶ Clear estimand, randomized design, prespecified monitoring, interpretable endpoints.
- ▶ Speed came from platform maturity and parallelization, not from abandoning rigor.
- ▶ Fast and careful were not opposites.

Good design $\implies$ credible speed

What worked well III: learning while treating, the RECOVERY trial

- ▶ The RECOVERY¹ (Randomised Evaluation of COVid-19 thERapY) trial launched March 2020 — multiple arms, shared control, arms added and dropped as evidence accumulated.
- ▶ Dexamethasone found to reduce 28-day mortality by one third in ventilated patients — result released in June 2020.
- ▶ Pragmatic randomization inside routine clinical care made enrollment fast without sacrificing rigor.
- ▶ Sharp estimand + clean design = an actionable answer the moment it arrived.

Estimated to have saved over one million lives globally.

¹The RECOVERY Collaborative Group includes Thomas Jaki, UofSC Statistics PhD graduate.

What the stress test revealed

- ▶ **Inconsistent reporting.** Case counts, hospitalizations, and deaths were defined and reported differently across states, countries, and time.
- ▶ **Missingness and changing definitions.** Who counted as a COVID death changed. Who got tested changed. The data we were analyzing was a moving target.
- ▶ **Causal inference under policy variation.** Lockdowns, mask mandates, and school closures were implemented differently everywhere — making it nearly impossible to cleanly estimate what worked.
- ▶ **Uncertainty communication.** We knew how to quantify uncertainty. We did not know how to communicate it to a public that wanted a single number and a clear answer.

*These were not new problems.
COVID just made them impossible to ignore.*

From genome to vaccine in less than a year

- Jan 10, 2020 Chinese researchers post the SARS-CoV-2 genome sequence publicly.
- Jan 13, 2020 Moderna and NIH finalize the mRNA-1273 vaccine sequence — 72 hours later.
- Mar 16, 2020 First human trial begins.
- Dec 11, 2020 FDA emergency use authorization for BNT162b2 (Pfizer–BioNTech).

Genome to authorization: less than a year.

Previous fastest vaccine (mumps): 4 years.

What we did well

Vaccine sequence in 72 hours

Rigorous trials in months

Clear, replicable results

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What we could not do

Get people to take it

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We solved the problem we knew how to solve.

We need to prepare for the problem we didn't.

Who did not take the vaccine and why

- ▶ Hesitancy was not uniformly distributed — it clustered by political affiliation, religion, race/ethnicity, and rural versus urban location.
- ▶ The most commonly cited reasons:
 - ▶ Concern about side effects and long-term safety.
 - ▶ Distrust of government and pharmaceutical companies.
 - ▶ Belief that the vaccine was developed too quickly.
 - ▶ Low perceived personal risk from COVID-19.
- ▶ Primary care physician recommendation was one of the strongest predictors of uptake.
- ▶ Hesitancy was more predicted by trust in institutions than by education level or access to information.

One of the most important scientists of the pandemic:

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What we are not trained to do

- ▶ Statisticians are trained to communicate to other statisticians.
- ▶ Our papers are written for journals — not for school boards, city councils, or the evening news.
- ▶ We have no formal training in risk communication, yet we are the ones who understand risk best.

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*We have optimized for depth.
We have underinvested in reach.*

What a statistics course actually teaches

- ▶ Not just t -tests and p -values.
- ▶ How to ask whether a sample is representative.
- ▶ How to reason about risk.
 - ▶ How to reason about risk that is probabilistic, not certain.
- ▶ How to update beliefs when new evidence arrives.
- ▶ How to recognize when a chart is lying to you.

*These are not just statistical skills.
They are civic skills.*

← Tweet



Dare Obasanjo
@Carnage4Life

You can teach an entire semester of how to lie with statistics with the y-axis of this chart.



7:25 PM · Apr 4, 2020 · Twitter for iPhone

14.2K Retweets 66.2K Likes

The department as a public health intervention

- ▶ 40 years. If the department has averaged 1,000 students per year, that is 40,000 people carrying statistical thinking into South Carolina communities.
- ▶ These are nurses, teachers, journalists, lawyers, real estate agents, business owners — not statisticians.
- ▶ Every one of them sat in a room and learned to think more carefully about evidence, risk, and uncertainty.
- ▶ The benefit to society has been tremendous.

The fight is not over



AJPH CALL FOR PAPERS
AJPH Special Section
Global Antiscience Phenomenon and Its Impact on Public Health

Submission Due Date: April 30th, 2026.

Social and political opposition to scientific developments in public health is not a new phenomenon, but it seems to have become more active, or at least more visible, in recent years, particularly regarding the global response to the COVID-19 pandemic. Antiscience activism has extended to political discourse and policymaking, threatening the development of

The next 40 years

- ▶ Research will always matter — new methods, new theory, new applications.
 - ▶ Clearly, there are things we don't yet understand.
- ▶ But one of the next frontiers is reach... through new applications and to the public.
- ▶ Four directions:
 - ▶ Training for graduate students on the effective communication of statistics for people of all levels.
 - ▶ Deeper engagement with K–12 education.
 - ▶ More emphasis on “reach” in T&P evaluations and awards.
 - ▶ Public-facing work: op-eds, expert testimony, scientific communication.

Depth got us here. Reach is what comes next.

40 years. 40,000 people. What comes next.

*The department's greatest contribution to South Carolina
is not its publications.*

*It is the 40,000 people who left its classrooms
thinking more carefully about the world.*

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*It is the 40,000 people who left its classrooms
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*The next 40 years is about reaching 40,000 or 100,000 more —
and making sure they are ready for the next stress test.*

Thank you

Questions and discussion welcome.

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`alexmcclain.github.io`

